USV Lab 1

System Identification

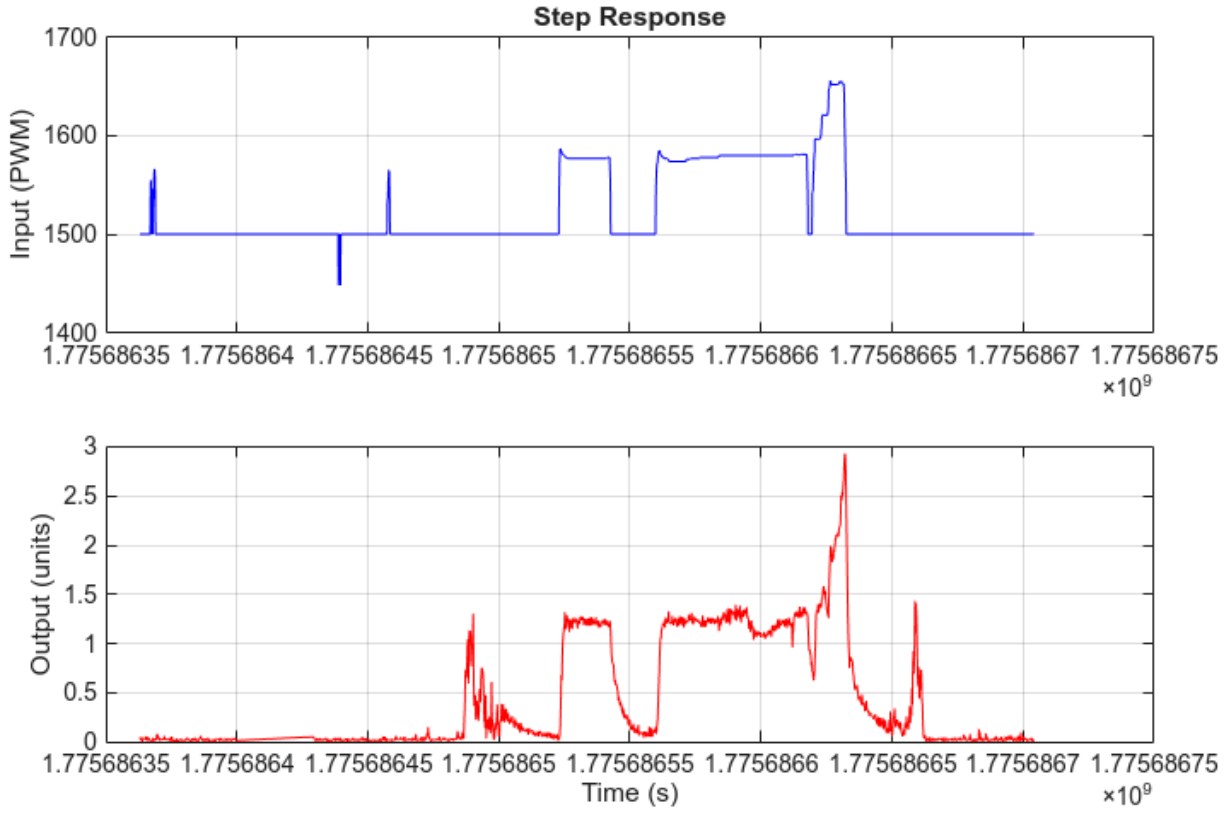
Find simple transfer function models
to approximate the steering and yaw response

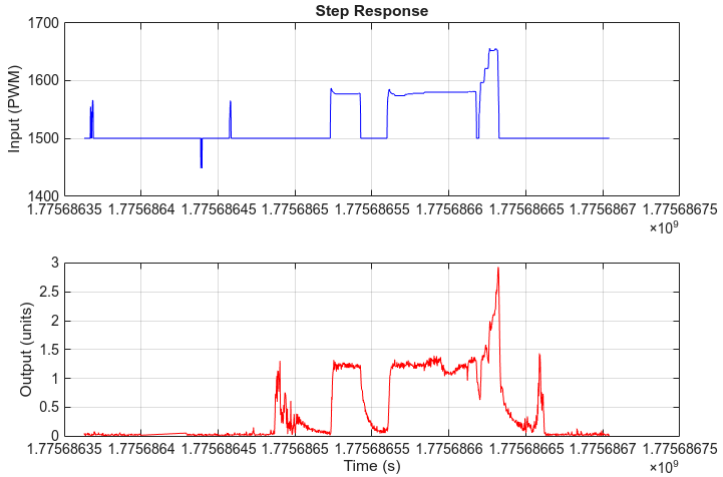
Objectives

Given the experimental dataset

Create two first-order transfer function models (surge and yaw-rate) to approximate the USV dynamics for use in feedback control design.

These models allow us to simulate the USV in MATLAB.





Surge plant	$G^*_{\text{surge}}(s) = \dfrac{K^*_u}{\tau_u s + 1}$
Yaw-rate plant	$G^*_r(s) = \dfrac{K^*_r}{\tau_r s + 1}$
Heading plant	$G^*_{\text{yaw}}(s) = \dfrac{K^*_r}{s(\tau_r s + 1)}$

- **Load data** — preprocessed ArduPilot log containing actuator commands and vehicle response.
- **Select channel** — choose one input/output pair (e.g. throttle → speed, rudder → yaw rate).
- **Isolate a step** — manually clip a single clean step response and zero the time axis.
- **Scale input** — convert PWM to normalized effort (±1).
- **Initial estimate** — read DC gain and time constant off the plot by inspection.
- **Curve fit** — refine parameters with lsqcurvefit to match the experimental data.
- **Result** — identified first-order model

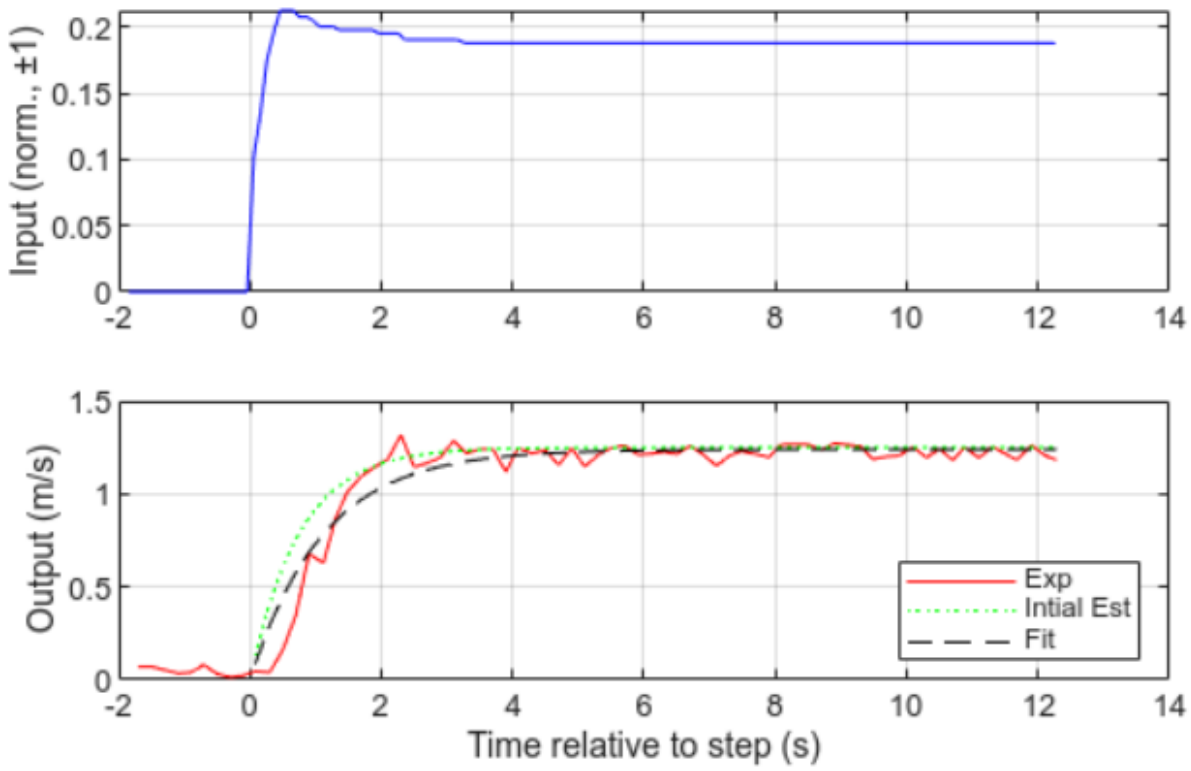
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Local minimum possible.

lsqcurvefit stopped because the final change in the sum of squares relative t
its initial value is less than the value of the function tolerance.

<stopping criteria details>


K_fit = 2.1369

tau_fit = 1.1130
```



Model: General Form

$$G(s) = \frac{K_{DC}}{\tau s + 1}$$

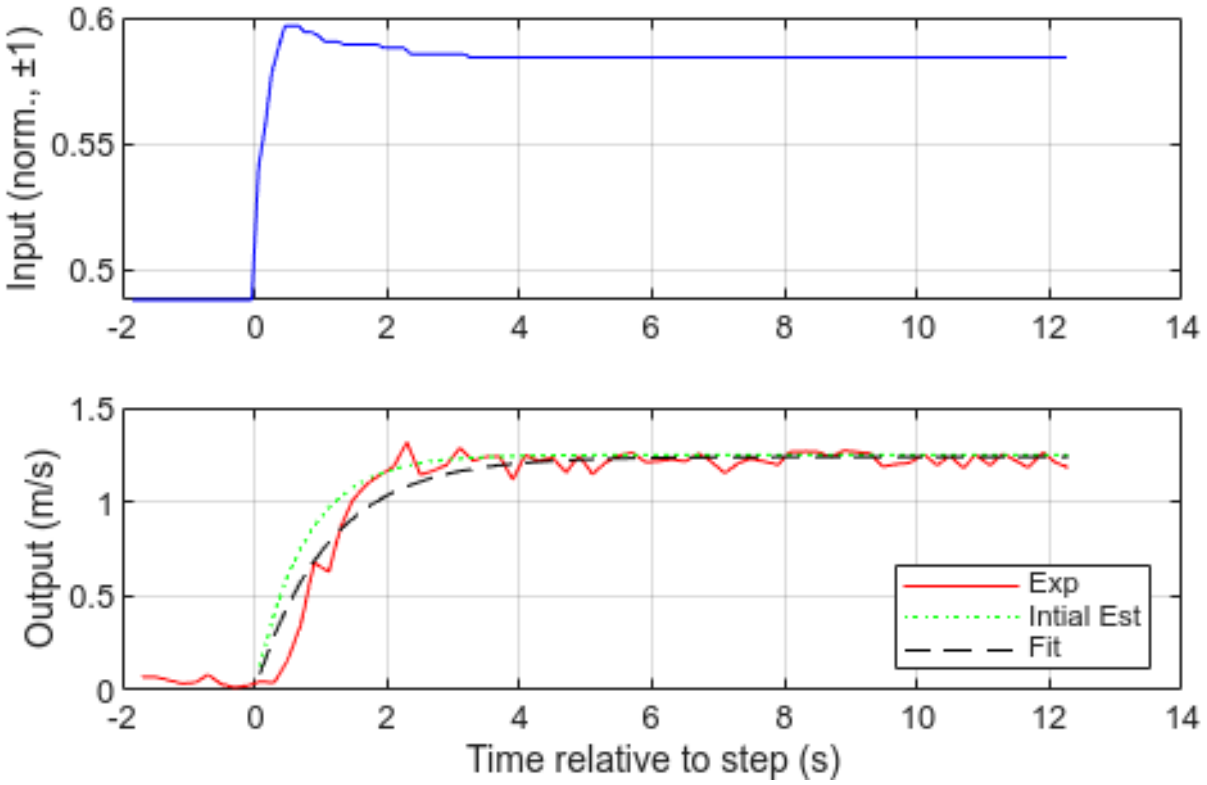
Model: System Identification

$$G(s) = \frac{2.1369}{1.113 s + 1}$$

Lab 1: Presentation

- Presentation
  - Title with team name and member names
  - Two slides. Each slide has
    - Final step response data plot with correct units.
    - Table with model parameters
    - Transfer function expression
  - May use the following slides as a template.
  - Submit as pptx file.
    - Shared copy, but each person submits via Sakai

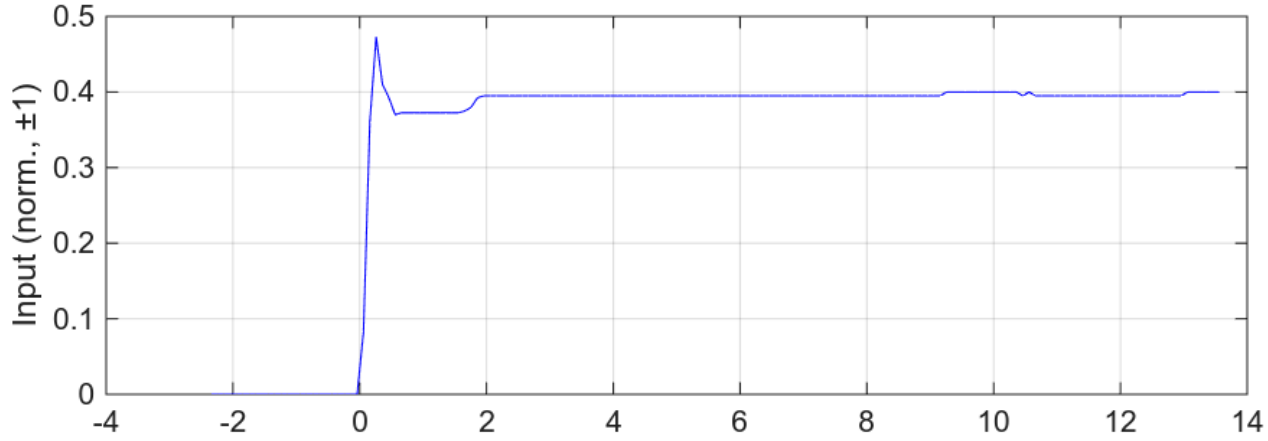
USV Surge Model Identification

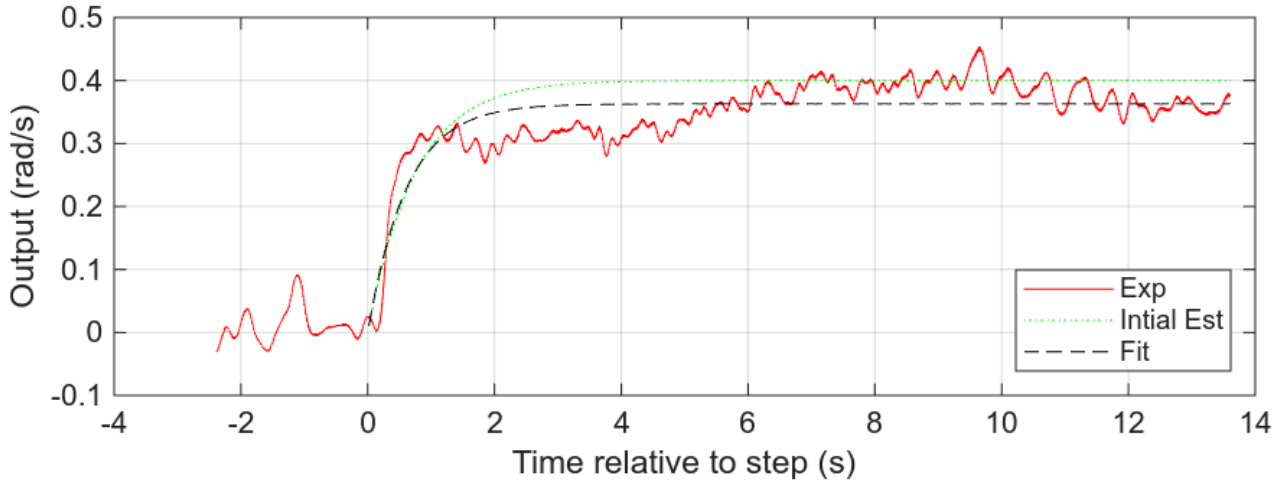


Model Parameter	Value	Units
DC Gain (Kdc)	2.14	m/s
Time Constant (tau)	1.11	s

$$G_{\text{surge}}(s) = \frac{2.14}{1.11\,s + 1}$$

USV Yaw-Rate Model Identification



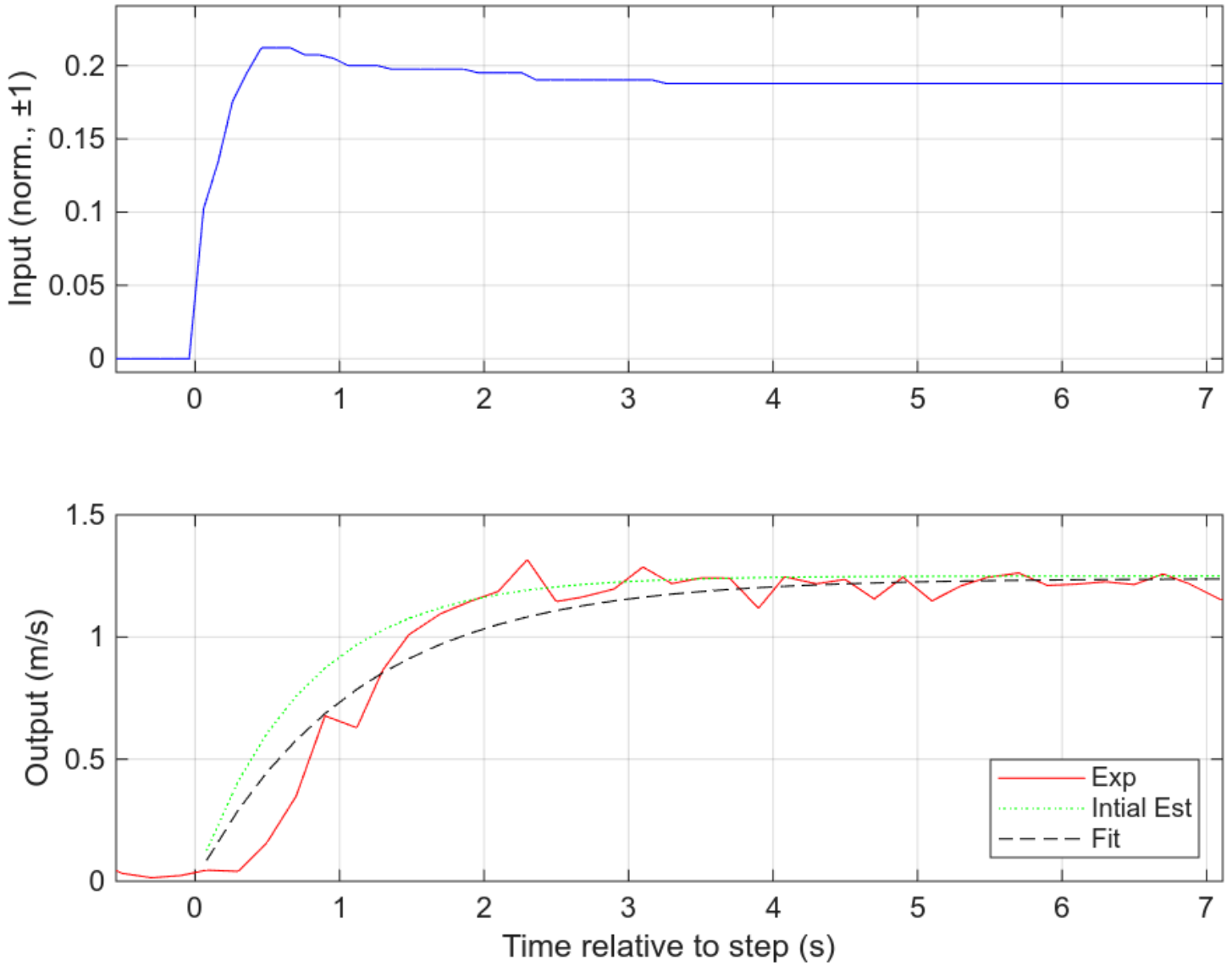


Model Parameter	Value	Units
DC Gain (Kdc)	0.52	rad/s
Time Constant (tau)	0.61	s

$$G_{\text{yaw-rate}}(s) = \frac{0.52}{0.61\,s + 1}$$

Model Limitations

- There are always "unmodelled dynamics". The challenge is knowing how much is sufficient.



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